

Technical Document #20: Data Engineering - Comprehensive Overview

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Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Data pipelines automate the movement and transformation of data between systems.
- Data warehouses store structured data optimized for analytical queries.
- Data lineage tracks the origin and movement of data through systems.